

CERTIFIED ASSESSMENT ASSIGNMENT 2: RESPONSIBLE AI ETHICS AUDIT

Participant Name: Muhammad Ashraf | **AGAIF Reference Code:** MY-411 | **Module / Task:** Module 1
Session 2 – GeoAI Ethics Audit Exercise (CA2) | **Target System Evaluated:** AI-Based Geospatial Flood
Warning System | **Assessment Date:** August 11, 2026

Evaluation of AI-Based Geospatial Flood Early Warning System

Hands-On Activity 6 — Ethics Audit Exercise

Task: evaluate an AI-based flood warning system against each of the seven dimensions using the traffic-light approach:

- Green** = Acceptable — evidence exists that this dimension is handled.
- Yellow** = Needs improvement — partially handled, or handled without evidence.
- Red** = High risk — unaddressed, and could cause real harm.

No.	Dimension	Green (Acceptable)	Yellow (Improve)	Red (High Risk)	Justification, Identified Ethical Risks, Stakeholder Impact & Mitigation Measures
1.	Fairness		⚠ IMPROVE		Current Condition & Justification: Model accuracy is predominantly tested in heavily gauged urban river basins (e.g., Klang Valley). Rural, indigenous (Orang Asli), and informal settlement river catchments suffer from sparse gauge density, resulting in spatial data under-representation. Identified Ethical Risk: Algorithmic Bias & Spatial Disparity — Higher false-negative rate (missed flood alerts) in low-gauge rural zones. Impact & Stakeholders: Rural agricultural communities and low-income informal settlers face unannounced flood inundation, leading to severe property destruction, livestock loss, and potential casualties. Mitigation Measures: 1) Integrate Sentinel-1 SAR satellite water masks to supplement low-gauge rural basins. 2) Deploy low-cost IoT stream gauges in remote hot-spots. 3) Re-weight training loss to equalize spatial prediction sensitivity across urban and rural catchments.

No.	Dimension	Green (Acceptable)	Yellow (Improve)	Red (High Risk)	Justification, Identified Ethical Risks, Stakeholder Impact & Mitigation Measures
2.	Privacy			✗ HIGH RISK	Current Condition & Justification: The system ingests and centrally stores exact household GPS coordinates, personal phone numbers, and residential address records paired with flood-vulnerability scores without anonymization or encryption. Identified Ethical Risk: Privacy Infringement & Unsanctioned Surveillance — Central storage of raw, unencrypted PII exposes sensitive residential location data. Impact & Stakeholders: Property owners and residents risk data leakage, unauthorized commercial exploitation, real-estate devaluation, or predatory insurance rate discrimination. Mitigation Measures: 1) Enforce spatial k-anonymity and aggregate public alert dispatch to 250m H3 hexagonal spatial grids rather than raw household points. 2) Implement end-to-end PII encryption and Role-Based Access Control (RBAC). 3) Enforce strict data minimization and purge transient contact telemetry post-alert delivery.
3.	Transparency		⚠ IMPROVE		Current Condition & Justification: The AI system outputs opaque, black-box probability scores (e.g., 'Risk = 0.88') without explaining the underlying spatial drivers or physical triggers. Identified Ethical Risk: Opacity & Explainability Deficit — Emergency managers and residents receive unexplained numbers without physical context. Impact & Stakeholders: District emergency officers (BOMBA, SMART) experience decision hesitation during evacuations, while residents lack clarity on flood severity drivers (e.g., dam release vs. extreme rainfall). Mitigation Measures: 1) Integrate Explainable AI (XAI) feature attribution maps (SHAP/Integrated Gradients) into emergency dashboards. 2) Convert probabilities into natural-language alerts (e.g., 'High risk driven by 120mm/hr upstream rainfall combined with 16:00 high tide'). 3) Publish an open Model Card detailing dataset boundaries, accuracy limitations, and operating constraints.

No.	Dimension	Green (Acceptable)	Yellow (Improve)	Red (High Risk)	Justification, Identified Ethical Risks, Stakeholder Impact & Mitigation Measures
4.	Accountability			✗ HIGH RISK	Current Condition & Justification: No formal governance framework or legal RACI matrix exists defining explicit liability between GeoAI developers, hydrologists, and public emergency response agencies in the event of system failure. Identified Ethical Risk: Diffusion of Responsibility & Unclear Liability — Absence of a designated entity held accountable for false positives (costly panic) or false negatives (unwarned disasters). Impact & Stakeholders: Disaster response agencies face legal vulnerabilities and operational deadlock, while affected flood victims are left without legal recourse or formal redress mechanisms. Mitigation Measures: 1) Formulate an Inter-Agency Memorandum of Understanding (MoU) establishing that the AI operates strictly as decision support, while human Civil Defense Directors retain final decision authority. 2) Mandate an independent Post-Event AI Performance Audit Committee. 3) Maintain an immutable, tamper-evident audit log of all predictions and human overrides.
5.	Safety & Reliability		⚠ IMPROVE		Current Condition & Justification: The model lacks real-time confidence metrics, Out-of-Distribution (OOD) monitoring, and automated fallback controls when input sensors (telemetry/radar) become corrupted or degraded during extreme storms. Identified Ethical Risk: System Degradation & Misleading Precision — Risk of confident but dangerously incorrect predictions during unprecedented climate events exceeding historical training data. Impact & Stakeholders: Emergency responders and vulnerable populations risk relying on degraded AI outputs, resulting in misallocated rescue boats, missed evacuations, or fatal delays. Mitigation Measures: 1) Deploy real-time OOD distance monitors and sensor data sanity checks. 2) Display explicit spatial confidence bands (High/Medium/Low) on responder maps. 3) Implement automated fail-safe fallback to deterministic physics-based hydrological models (HEC-RAS/SWMM) when ML uncertainty exceeds safety thresholds.

No.	Dimension	Green (Acceptable)	Yellow (Improve)	Red (High Risk)	Justification, Identified Ethical Risks, Stakeholder Impact & Mitigation Measures
6.	Sustainability	✓ ACCEPTABLE			Current Condition & Justification: The system utilizes lightweight, optimized spatial inference pipelines (pruned LightGBM/U-Net) running on modest, energy-efficient cloud infrastructure with minimal carbon footprint. Identified Ethical Risk: Low Environmental Risk — Operational energy usage and computational costs are well-managed and sustainable. Impact & Stakeholders: Positive long-term financial and environmental viability for municipal disaster budgets. Mitigation Measures: 1) Maintain green computing standards by hosting cloud retraining pipelines in regions powered by renewable energy. 2) Utilize incremental delta-training updates rather than full retraining runs to minimize compute overhead.
7.	Human Oversight	✓ ACCEPTABLE			Current Condition & Justification: Designed with a mandatory Human-in-the-Loop (HITL) protocol where automated AI alerts require dual sign-off from duty hydrologists before public broadcast, supported by community crowd-sourced ground-truth reporting. Identified Ethical Risk: Adequately Controlled Risk — Human verification effectively prevents unchecked automated errors. Impact & Stakeholders: Protects the public from false panics while empowering local residents to submit real-time ground-truth flood reports. Mitigation Measures: 1) Conduct regular scenario training for duty officers to prevent 'automation bias' (blindly trusting AI) or 'alert fatigue'. 2) Maintain community ground-truth reporting loops to validate and calibrate AI prediction grids in real time.

Tip: for every Yellow or Red, write one concrete mitigation in the Notes column — this table then becomes the seed of your project's Ethical Risk Register (Session 5, Activity 5).

Worked hints for the flood-warning scenario:

Fairness — is accuracy tested in low-gauge areas (Yellow if unknown)?

Privacy — does it store exact household coordinates it doesn't need (Red if yes)?

Transparency — do alerts explain themselves (Yellow if just a score)?

Accountability — is anyone named (Red if no)?

Safety — does it show confidence (Yellow if silent)?

Sustainability — is the model right-sized (usually Green for a small model)?

Human Oversight — can residents report a missed flood (Green if yes)?

OVERALL DEPLOYMENT RECOMMENDATION & CONCLUSION

FINAL AUDIT VERDICT: **PROCEED SUBJECT TO SPECIFIED CONDITIONS** (Conditioned Approval)

Executive Summary & Decision Rationale:

The proposed AI-Based Flood Early Warning System exhibits significant potential value for public safety, disaster preparedness, and rapid emergency response. However, full regional deployment must NOT proceed in its current un-mitigated state due to two (2) critical RED high-risk dimensions (Privacy and Accountability) and three (3) YELLOW dimensions (Fairness, Transparency, and Safety & Reliability).

Mandatory Pre-Deployment Conditions (Prerequisites for Pilot Launch):

- **1. Privacy Remediation (RED Risk):** Eliminate central storage of raw household GPS coordinates and unencrypted PII. Enforce spatial k-anonymity (250m H3 hex grid aggregation) and complete an independent Privacy Impact Assessment (PIA).
- **2. Legal Accountability Framework (RED Risk):** Ratify an Inter-Agency Memorandum of Understanding (MoU) formally declaring that the AI operates strictly as decision support, while human Civil Defense Directors retain sole legal evacuation authority. Establish an immutable audit logging pipeline.
- **3. Safety & Fallback Controls (YELLOW Risk):** Implement real-time sensor sanity monitors and an automated fail-safe fallback to physics-based hydrological models (HEC-RAS) when ML uncertainty is high.
- **4. Spatial Fairness Integration (YELLOW Risk):** Supplement sparse rural stream gauge telemetries with Sentinel-1 SAR satellite water masks and low-cost IoT gauges to eliminate urban-rural accuracy disparities.
- **5. Phased Pilot Rollout (90-Day Controlled Pilot):** Deploy the system exclusively as a limited 90-day pilot within a single controlled river basin (e.g., Selangor River catchment) under active human-in-the-loop validation before national expansion.

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